

Title	Authors	Year	Dataset	Application area	Method / Model	Metrics	Strengths	Limitations	Research gap	Relation to our work
FORS-EMG: A Novel sEMG Dataset for Hand Gesture Recognition Across Multiple Forearm Orientations	Rumman et al.	2024	19 subjects, 12 gestures, 8 channels, 3 orientations	Orientation-robust gesture recognition	LDA, SVM, KNN, 1D-CNN, RNN, LSTM	Highest reported cross-orientation F1-score was 88.58% using LDA with SNTDF.	Focuses on forearm orientation and provides baseline models	Small dataset, able-bodied subjects only	Better cross-subject and cross-orientation models are needed	Main dataset and baseline for the proposed CNN study
putEMG: A Surface Electromyography Hand Gesture Recognition Dataset	Kaczmarek et al.	2019	44 subjects, 8 channels, 2 sessions	Prosthetic control and wearable HMI	LDA, QDA, kNN, SVM with RMS, Hudgins and Du features	SVM/RMS achieved about 89% precision.	Large cohort, repeated sessions, raw signals and video-based labels	No deep-learning benchmark, controlled posture, able-bodied subjects only	Deep learning and stronger channel-based feature learning are needed	Supports channel attention and global-local feature extraction
Enhanced Performance for Multi-Forearm Movement Decoding Using Hybrid IMU-sEMG Interface	Shahzad et al.	2019	10 healthy subjects, 6 hand gestures, 4 sEMG channels, 2 IMUs	Prosthetic control and hand gesture recognition	LDA, SVM, Hybrid IMU-sEMG, TD & TD-POS features	Highest accuracy was 98.7% using SVM with TD-POS features (dynamic training).	Combines IMU and sEMG for higher accuracy and robust arm-position recognition	Small dataset; only healthy subjects; no deep learning model	Deep learning and cross-subject validation are needed for real-world applications	Supports sensor fusion and motivates our CNN-based hand gesture recognition model with improved feature learning
Machine Learning-based sEMG Signal Classification for Hand Gesture Recognition	Aarotale & Rattani	2024	GrabMyo, FORS-EMG	HGR, prosthesis, rehabilitation	fTDD, TSD, Wavelet, RF, SVM, KNN, Dilated CNN, CNN-LSTM	Accuracy, Precision, Recall, F1-score	Multiple datasets, Many models, Strong accuracy	Neutral orientation only, random split, No real-time test	Cross-subject, cross-orientation, real-time validation	Baseline feature-model comparison For proposed work
Quantifying Covariate Shift and Improving Electromyography Driven Gesture Recognition with Calibration and Sample Selection	Hongquan Le, Marc in het Panhuis, Geoffrey M. Spinks, Gursel Alici	2024	11 subjects, 8 gestures, 7 channels (Multiple Limb Position); 10 subjects, 7 gestures, 14 channels (Ninapro DB6)	sEMG hand gesture recognition, prosthetic control, HMI	LDA, LDA-D, RI-DA, RI, NOI, NCC	<i>F1-score 88.58%</i> <i>SVM/RMS achieved about 89% precision."</i>	Improves recognition under covariate shift; introduces NOI and NCC metrics	Uses only LDA, no deep learning, evaluated on two datasets	Deep learning, automatic feature extraction, better cross-subject generalization are needed	Provides a baseline for our CNN-based hand gesture recognition model